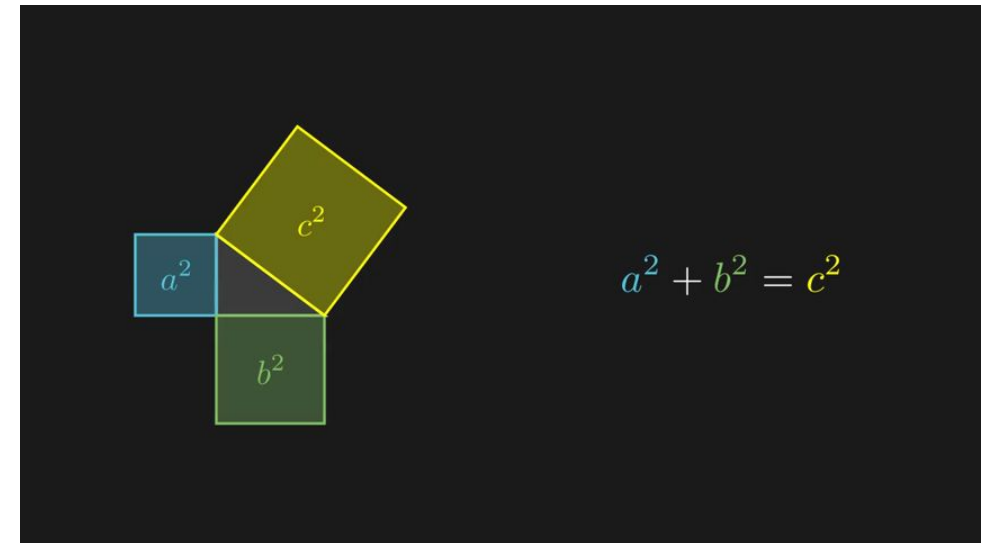


MANImation Studio

From one sentence to a narrated, 3Blue1Brown-style lesson video

Multi-agent AI system · Dhup Thumbadiya, Pallavi Pansuriya



A frame generated by the system: Pythagorean proof

Two ways to make a 1-minute math animation today

Traditional animation software

After Effects · Blender · Adobe Animate

\$500 – 1000

estimated cost, 1-minute animation

1 – 2 weeks

time to produce it

Needs a professional animator, plus paid software licences

How every shape keyframed by hand; equations pasted in as images

Weak spot accuracy depends on the animator; changing one number means redoing the shot

Code libraries (e.g. Manim)

Manim · Python · LaTeX: free and open source

\$50 – 200

estimated cost, 1-minute animation

1 – 3 days

time to produce it

Needs subject knowledge and coding skill (Python, LaTeX)

How animation written as code; the engine computes every position exactly

Weak spot most teachers can't program, and debugging takes most of the time

The gap: code libraries make exact animations cheap, but only for people who can program. MANImation Studio lets the AI write that code: our real run cost avg **\$0.1 for a 68-second narrated video**.

Cost estimates: typical freelance rates for 2D explainer animation and for a Python developer's time.

Why not just ask an AI video generator?

Video models like Sora or Veo learn what videos look like, not what the math means. They predict pixels, so:



Equations come out garbled

There is no LaTeX inside a video model. “ x^2 ” is just a shape it tries to draw, so symbols and labels get mangled.



The math isn't exact

A right angle drifts, a curve wobbles between frames. Nothing checks that what is drawn is actually true.



You can't fix one detail

Change one number and the whole clip is regenerated, and it looks different. Clips are also short and not synced to narration.

AI video model

Prompt → guesses pixels, frame by frame → a video that only looks right

vs

MANimation Studio

Prompt → AI writes Manim code → the engine draws it with exact geometry and LaTeX

Our choice: the AI writes code, and a real math engine draws every frame: exact, checkable, editable.

Our idea: describe it, review it, get the video

1

Describe

Type one sentence and choose a color theme, the audience and a budget.

2

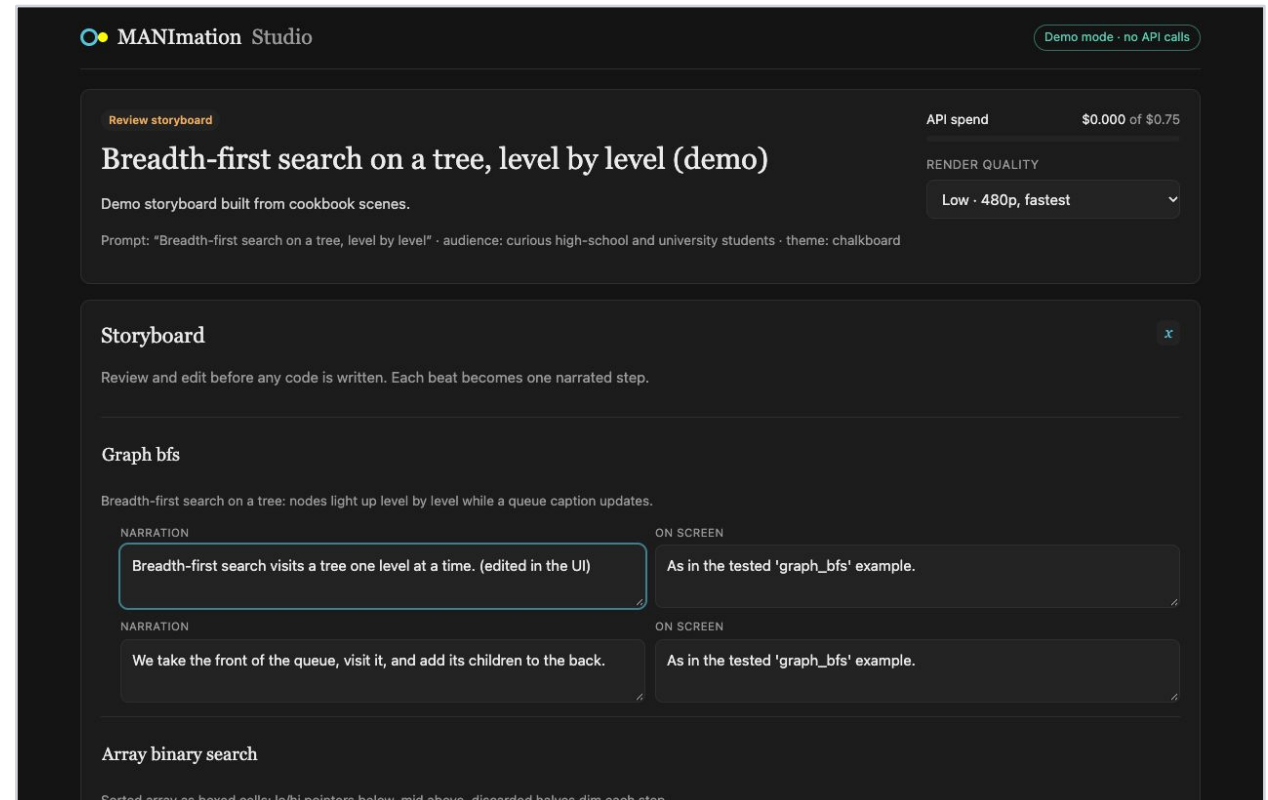
Review

Read the storyboard the AI planned and edit any narration or visual before code is written.

3

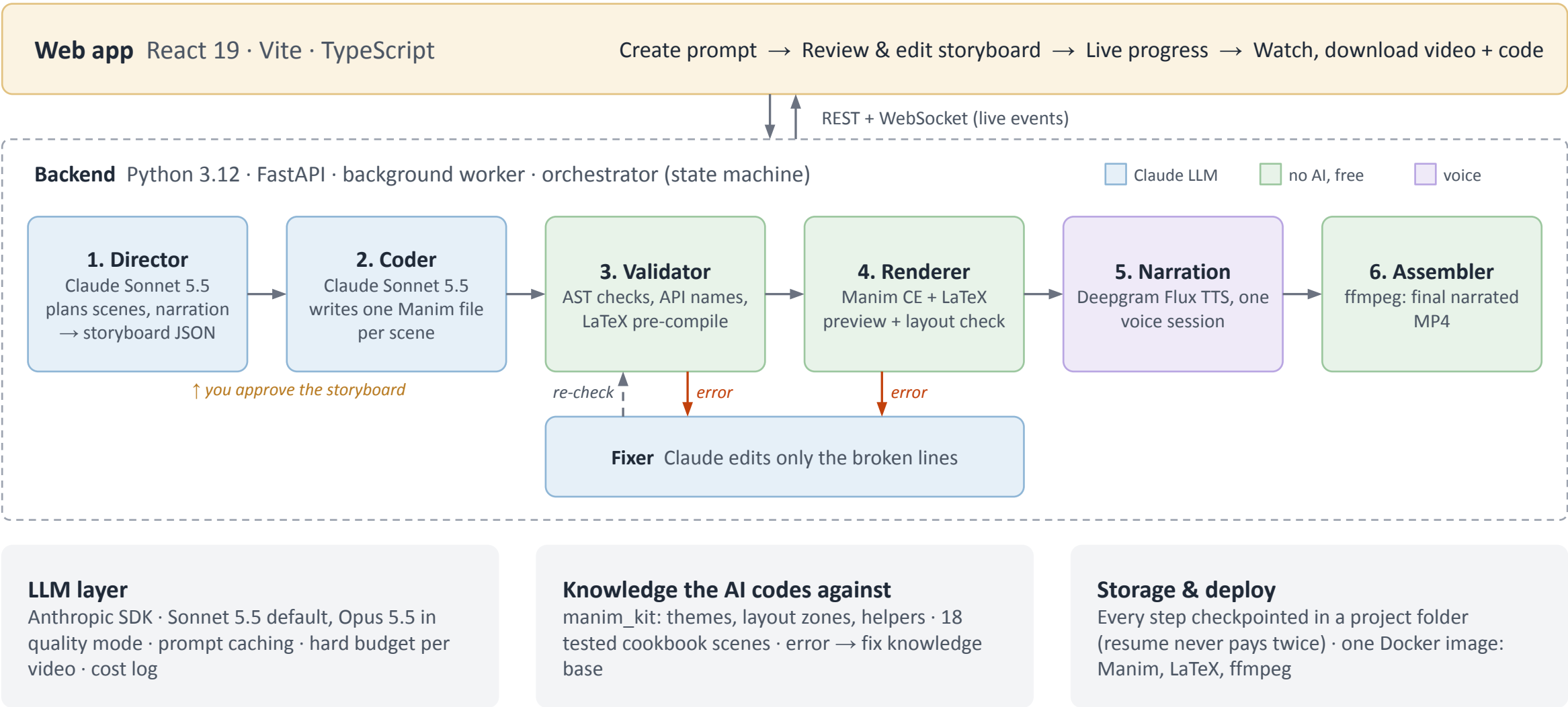
Receive

Watch and download the narrated video, plus the Manim code for every scene.

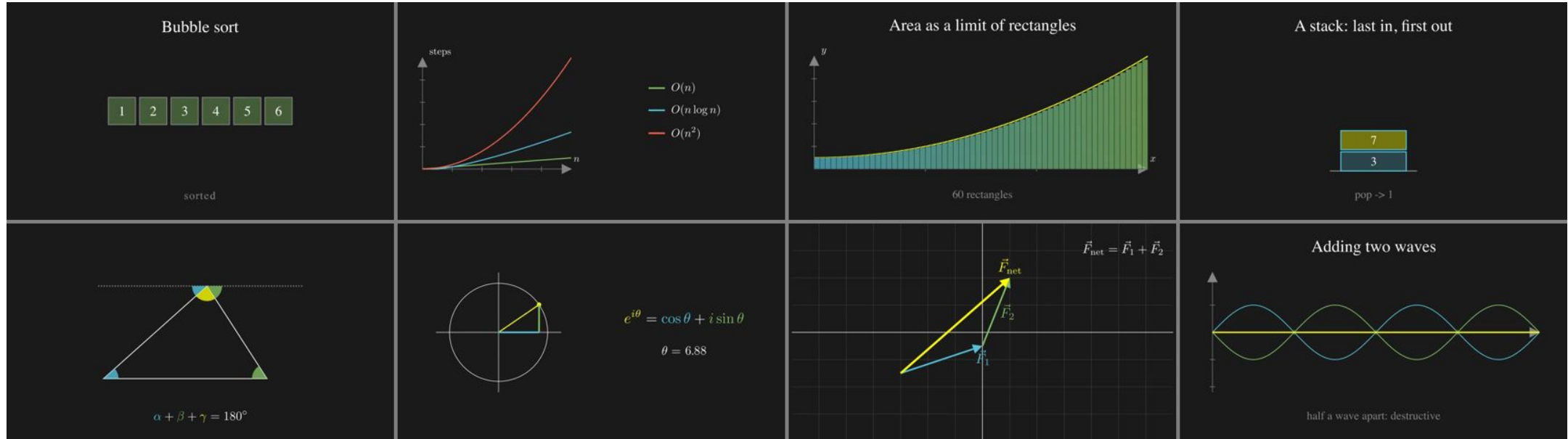


The storyboard review screen of the web app

Architecture: how everything fits together



What it makes today, and what comes next



Frames from scenes the system renders: algorithms, calculus, physics, geometry, linear algebra

Working today

- Full web app: create, review, live progress, request changes
- Narrated video in four themes, plus the source code
- Runs offline in demo mode and in one Docker container

Next

- Visual critic: an AI that reviews rendered frames
- Larger evaluation on 15 math, physics and CS topics
- Slide-clip mode for lecture slides; more voices and languages